

An Integrated Core-and-Cluster Framework for Teaching Physical Examination Aligned with Clinical Reasoning

Background

Early medical curricula commonly teach physical examination as a comprehensive head-to-toe routine. While foundational, this is time-intensive, inconsistently reflected in routine practice, and misaligned with how clinicians select maneuvers in response to a specific problem. Learners struggle to distinguish elements performed universally from those applied selectively, which limits the integration of examination with clinical reasoning. The gap is documented and is not confined to beginners: senior students have been shown to know how to perform maneuvers without reliably knowing when to apply them.

Summary of Work

We implemented a Core + Cluster physical examination framework, adapted from the model described by Gowda and colleagues, within an early clinical skills curriculum at one medical school. Five faculty who lead clinical skills teaching, the preclerkship and clerkship curriculum, standardized patients, and simulation carried out the work.

Three decisions fell to us. First, which maneuvers form the core: we began from the 37-maneuver national consensus core and reduced it to 24 for a first-year learner, relocating seven maneuvers into clusters rather than discarding them. Three universal items (hand hygiene before and after patient contact, and draping) are performed at every encounter and, as in published cores, are counted separately. Second, which presenting problems draw each of eight domain-based clusters: cardiac, pulmonary, gastrointestinal, neuro-vestibular, musculoskeletal, HEENT, genitourinary, and endocrine/systemic. Third, that a presentation draws one cluster rather than several.

Every maneuver in the core and in every cluster carries written full and partial performance descriptors with observable criteria. These descriptors were our school's existing scoring standard and were adopted unchanged, so the framework altered which maneuvers a student performs, not the standard against which performance is judged.

Students learned the full head-to-toe examination in the first semester, to fluency. Core and Cluster was introduced in the second semester, with each cluster timed to the organ-system course running in parallel, so that the relevant pathophysiology was current rather than recalled. Cluster sessions covered the schema for the presentation and illness scripts for four to five competing diagnoses. Students practiced with standardized patients and in differential diagnosis sessions on paper cases, where they were asked what they thought was happening, what they would examine and what they were looking for, and how the findings would change their thinking.

Summary of Results

The framework comprises one core examination of 24 maneuvers in seven sections, eight clusters, and presenting-problem pathways. Clusters add between 3 and 20 maneuvers beyond the core, averaging 8, together with 40 optional add-ons requiring equipment, a chaperone, or a wider scope of practice. Musculoskeletal is organized by region, so a presentation draws only the group for the joint involved. Clusters are applied singly: each encounter is the core plus the one cluster indicated.

Before implementation we measured our own students. On a headache case, the neurologic maneuvers the presentation called for were not performed at all in 22% of instances, against 7% for maneuvers performed on every patient. On a cardiopulmonary case with the same students, nearly all attempted jugular venous pressure but only 39% performed it to standard. Selection did not follow the case, and a frequently attempted maneuver was frequently done poorly.

To interpret any subsequent change we first established how much these scores move on their own. Across four standardized-patient cases given to three consecutive classes taught head-to-toe, and 37 maneuvers, the difference between the best and worst year averaged 8.9 percentage points with no change in teaching. We treat that as the margin of ordinary variation between classes.

One class taught under the framework was then compared with those three head-to-toe classes on a cardiopulmonary standardized-patient encounter, given at the same point in training under the same instruction. The comparison was restricted to the four indicated maneuvers scored with word-for-word identical criteria in both years. Complete performance rose from 72.3% to 88.2%, a gain of 15.9 points, and each of the four differences fell outside the margin of ordinary variation.

Discussion and Conclusion

Separating universal maneuvers from context-specific additions aligns examination teaching with how clinicians work, while preserving a consistent baseline for every patient. Because the performance descriptors were fixed and written beforehand, a difference in scores between classes cannot reflect a change in standard; it reflects a change in what students did.

The only prior comparison of these two curricula measured a global outcome more than two years after instruction, found no difference, and attributed the null to washout of the teaching effect. Measuring proximally, at the point of instruction and on the maneuvers a presentation indicates, may be where an effect is detectable. Our result is consistent with that reading rather than in conflict with it.

Four limitations bound what this can show. Core and Cluster was added in the second semester and all four classes shared the same first semester, so this does not test a curriculum that begins with the core; it also makes the test conservative, since every class already knew how to perform the four maneuvers and the difference concerns whether they performed them for this patient. We compared separate classes rather than randomizing, so this is an association rather than evidence of cause. We measured what students did and not why, and scored completeness rather than accuracy: a randomized trial of hypothesis-driven versus screening neurologic examination found the hypothesis-driven approach detected more abnormalities but with lower specificity, and our outcome cannot speak to that trade-off. Finally, the assignment of maneuvers to clusters was agreed by our faculty and has not been tested elsewhere.

A cohort taught the core first sits the same examination next year, and a measure of reasoning quality rather than maneuver completeness is the necessary next step. What we report here is one school's

implementation and some early data, offered in the terms the model's own authors called for: comparison against historical controls.

This abstract accompanies the presentation delivered at AMEE 2026 (Session 4J, 24 August) and includes the comparison reported there. The abstract published in the conference proceedings was written before those data were available and differs in its Results and Conclusion. The Core + Cluster model is described by Gowda and colleagues; this is one school's adaptation.